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MMDBHDSNet: A MultiModal Dual-Branch Hazard-Discriminative Framework for Real-Time Medical Waste Segmentation and Risk-Aware Analysis

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Abstract

Medical waste management has been a thorny issue in hospital facilities, with improper segregation still presenting concrete risks to the sanitation sector and the local communities. Automated vision systems provide stable, 24 hour detection but all available solutions have three basic shortcomings: lack of class coverage, lack of hazard level reasoning and lack of a method to measure the uncertainty of predictions. This paper introduces MMDBHDSNet, a MultiModal Dual-Branch Hazard-Discriminative Segmentation Network that performs real-time pixel-level detection and risk-stratified classification of medical waste across 38 clinically defined categories. Two parallel decoder branches share a common BiFPN encoder — one optimised for instance segmentation quality, the other encoding hazard semantics — and exchange information through a Cross-Branch Fusion Gate that prevents gradient interference. A spatial contamination risk module aggregates co-occurrence relationships among detections into a scene-level score, and a Monte Carlo Dropout block estimates epistemic uncertainty per detection. To support training and evaluation, seven independently annotated source datasets were merged through a six-stage engineering pipeline that resolved cross-corpus class-naming conflicts, removed 8,260 duplicate images via MD5 hashing and polygon-structure comparison, remapped all label IDs to a unified 38-class ontology, and rebalanced the train–validation–test partition to 70/20/10. The resulting benchmark contains 14,445 annotated images — the most thoroughly documented medical waste segmentation collection of its kind. On the held-out test partition, MMDBHDSNet achieves mAP@0.5 of 0.9474 and mAP@0.5:0.95 of 0.7613, with precision 0.951 and recall 0.9168. The deployment interface sustains up to 55 frames per second with live webcam inference, an EMA-smoothed risk trajectory graph, spatial heatmap overlays, and a three-tier escalation alert system suitable for supervised clinical integration.

Keywords: Medical waste detection, Dual-branch deep learning, Hazard-aware classification, Uncertainty estimation, Real-time instance segmentation, Contamination risk scoring, Healthcare AI, Monte Carlo Dropout

1. Introduction

Healthcare systems worldwide generate considerable hazardous waste daily. The World Health Organization estimates that clinical activities produce approximately 5.5 kg of hazardous material per hospital bed per day, including sharps, infectious biologicals, pharmaceutical residues, and radioactive substances [1]. Lack of proper sorting of waste at the time of disposal, which is still mostly done manually, leads to contaminated material entering municipal waste streams, higher needle injuries among collection workers, and chemical leachate harming the local eco systems [2, 3]. Human sorters are fallible: the result of repeated classification under time; poor training; packaging that appears the same of hazard levels all combine to produce fallible results, which no supervision system can remove completely.

Computer vision does not have those problems. It does not tire, processes a frame in milliseconds, and once deployed its behaviour is consistent. Yet the field has been slow to deliver a robust clinical-grade system for medical waste specifically, held back by three gaps: (i) existing datasets cover only five to fifteen waste classes, far below the true diversity of clinical streams; (ii) architectures treat medical waste as a generic detection task with no awareness that a needle misclassified as LOW-hazard has different consequences from a misidentified mask; and (iii) no published system includes uncertainty quantification — indispensable when a wrong call routes a sharp instrument to household waste. CNN-based classifiers reach above 90% accuracy on benchmarks like TrashNet, yet performance degrades sharply on out-of-distribution footage [4]. Instance segmentation work on Mask R-CNN improved localisation but medical waste evaluations remain on small private datasets [5, 6]. No prior work combines risk scoring, uncertainty quantification, and multi-source dataset harmonisation.

All three gaps are covered in this paper in five concrete contributions:

(1) **MMDBHDSNet —** a dual-branch encoder-decoder with a Cross-Branch Fusion Gate and hazard-weighted multi-task loss.
(2) **A 38-class benchmark —** 14,445 images from seven source datasets, processed through a six-stage deduplication and rebalancing pipeline.
(3) **A risk scoring contamination model -** a dynamic scene-level index is computed by using spatial co-occurrence, hazard weightings, and detection confidence.
(4) **Monte Carlo dropout uncertainty estimator -** flagging uncertain detections so that they can be reviewed by a human before decisions are made on whether to dispose of an object.
(5) **A real-time deployment interface -** live webcam inference up to 55 fps, smooth risk trajectory, heatmaps, and three-level notifications.

2. Related Work

Over the last decade, deep-learning waste detection has evolved in many aspects, starting as a relatively simple image-level classification task, it has over time evolved to become a much more advanced concept: complex scene understanding and instance segmentation. The early benchmark datasets such as TrashNet demonstrated to the research community that convolutional neural networks (CNNs) could reasonably classify waste into a few material classes [7]. Such architectures as ResNet-50, DenseNet-121, EfficientNet-B4, and MobileNetV2 came later, and these continued to push the classification accuracy above the 90% threshold - again, in highly controlled lab settings. The instant you leave those clean spaces into the muck of the real world occlusion, unpredictable scales, domain shift these methods start to falter. They were constructed to be closed-set and that was going to be their limit.

This was identified quite soon, which is why such object detectors as Faster R-CNN, YOLO, and SSD appeared. These techniques provided something the previous classifiers could merely not provide: the capability to localize and identify objects in unrestricted, disorderly scenes. The TACO dataset took the field one step further by introducing truly realistic, real-world waste problems having multiple objects present in a single image [4, 8]. Despite all that development, most published work remained concentrated on overall and general types of waste. The field of medical waste, its micromarked differentiation and hazard-related categorizations were mostly omitted. With the recent surveys proving once again that there is no standardized, large-scale, multi-class dataset of medical waste, this lack of such a dataset has since become one of the most urgent bottlenecks holding this whole field back [9, 10].

Another visual understanding was achieved through instance segmentation. Mask R-CNN (which also uses Faster R-CNN) has a parallel branch dedicated to mask prediction, enabling the model to both detect objects and simultaneously segment them at the pixel level [5]. It was incredibly effective, however, the reliance on region proposals made it have a genuine computational cost - such that it was hard to use in real-time environments. One-stage methods such as SOLO and SOLOv2 avoided this

issue altogether by using grid-based assignment methods, rather than region proposals, at the cost of only slightly reduced accuracy but with a correspondingly much faster inference [11]. Next came transformer-based models Vision Transformers (ViTs) and Swin Transformers that added a qualitatively new capability: the ability to learn long-range interactions and think about the context of a complete image [12, 24]. This proves to be of great importance when it comes to medical waste where the distinction between a contaminated and an uncontaminated object may be fine enough that local features alone may not be sufficient to pick it up.

The reason why feature pyramid networks (FPN) and related successors, in particular BiFPN, have become a staple of modern detection pipelines is that they can efficiently perform multi-scale feature fusion, and that is precisely what you want when objects of your dataset might be as small as a needle, or as large as a disposal bin [13, 16]. With that said, the architectures have a blind spot: they consider all classes as equally important. They do not possess an inherent idea of which mistakes to be more important than others.

This is where hazard conscious and safety critical vision enters into the picture as a unique and more and more significant direction of research. Familiar with misclassification is not a new concept in fields such as autonomous driving, industrial inspection and healthcare as certain mistakes are much more lethal than others. This has inspired cost-sensitive learning and loss formulations that are risk-conscious [14]. Confidence calibration and anomaly detection in medical imaging in particular have been integrated into diagnostic systems to enable them to be more reliable and understandable. However, in the case of medical waste management, hazard cognizant reasoning remains surprisingly unexplored in the literature. The existing systems are designed to maximize classes in the same way that putting a syringe into a non-hazardous object has an equivalent penalty to labeling a piece of cardboard.

Multi-task learning has been of interest as a mechanism to optimize interrelated goals, such as detection, segmentation and classification simultaneously. This is more parameter-efficiency theoretically with shared encoder-decoder architectures, which often end up with negative transfer, where the conflicting gradient across tasks leads to worse performance instead of improved performance [21, 27]. There have been numerous efforts to remedy this with task-specific branches or attention-based modulation of features, although a genuinely principled mechanism of meticulously regulated information sharing across tasks has yet to be solved.

Uncertainty estimation has become a niche issue instead of a near-necessity in safety-critical AI, especially in healthcare. Normal deep neural networks are deterministic - they give a prediction, and do not give any clue to the degree of confidence or uncertainty that that prediction has, which poses a hazardous possibility of giving overconfident false answers. Bayesian deep learning has been designed as a way to fill this very gap, making predictions in terms of probability distributions as opposed to point estimates. A popular and practical approximation introduced by Monte Carlo Dropout by Gal and Ghahramani utilized stochastic forward passes at inference time [15]. Alternatives include deep ensembles and evidential deep learning that are more likely to provide better calibration, but have higher computational cost [17, 18]. Uncertainty estimation in medical imaging has proven beneficial in enhancing clinical trust and becoming a practical human-in-the-loop workflow. But it has hardly been assimilated into real time waste management vision systems.

The other frontier that is worth considering is multimodal learning. The essence is to put the visual, textual and contextual cues together to formulate stronger and better generalizing models. This method has already shown significant improvements in biomedical AI [26]. Multimodal approaches have started being embraced successfully in fields such as radiology and pathology, yet environmental and waste management systems lag behind. The possibilities are actual - envisage stuff like adding metadata to the origin of a waste item or what the local environmental conditions resemble. Such a contextual cue might greatly enhance the forecast of hazards.

Lastly, dataset engineering should not be underestimated. Research after research has shown that the quality of data, the balance of classes and consistency of annotations often are more important than architectural refinement [19, 20]. Deduplication, label harmonization, and stratified splitting are not vegetable frills - they are fundamental to the construction of systems which do generalize. Nevertheless, medical waste literature has been mostly based on small, single-source datasets that are ill-equipped to enable scalable or reproducible studies.

To take a step back and review: object detection, segmentation, and uncertainty estimation have been realistically advanced. However, it is in the case of the unique requirements of medical waste management that the current solutions are still considerably wanting. The unmet needs are tangible and can be named: no large-scale standardized multi-class data exist in this field; no architectures that can be described to have any form of hazard-aware reasoning; no concept of multi-task learning that can be defined to have a principled exchange of inter-tasks; and no uncertainty estimation has been incorporated into real time systems running in this space. The suggested MMDBHDSNet framework has all four of these gaps in its design, and it unites the dual-branch learning, hazard-weighted optimization, and uncertainty-aware inference into a single and unified architecture.

3. Dataset Construction

Motivation. No publicly available benchmark covers all 38 waste categories at the annotation granularity instance segmentation requires. Seven independently annotated corpora were therefore merged: MedBin_dataset v1, v5, v6, and v7, plus New dataset versions nv2, nv3, and nv4. They were all photographed in various locations, such as hospital corridors, laboratory benches, outdoor disposal locations, and using varied imaging equipment, and this diversity of the source offers the overall representative quality of the final benchmark [19, 20].

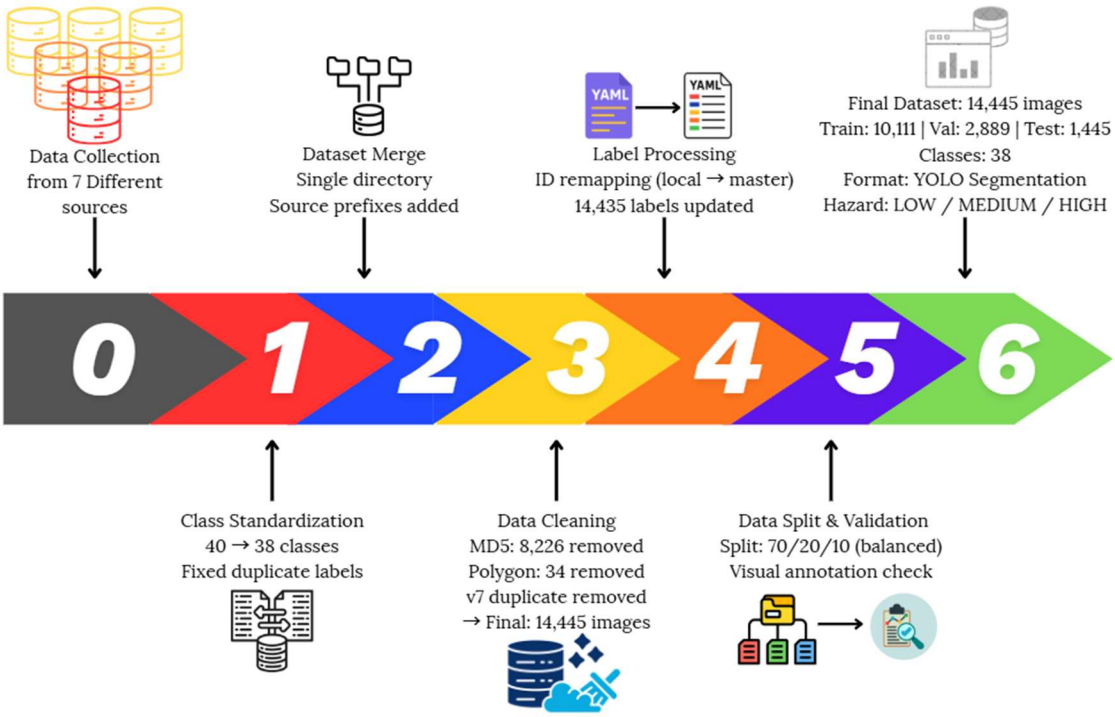

Figure 1: Multi-stage dataset construction and harmonization pipeline for the MMDBHDSNet framework.

Stage 1 — Class ontology. The source datasets were named independently and numbered by ID. Only one master ontology of 38 classes was locked. Two cross dataset inconsistencies were fixed: plastic medical gloves (in both v1 and v2) was standardised to medical gloves, and the annotation error medicine bottole (in v5) was fixed and combined into plastic medical bottle. The number of raw classes reduced to 38 by removing actual duplicates, and not information.

Stage 2 — Prefix-based merge. All of the images and their associated label files were renamed with a prefix of dataset-of-origin (e.g. v5_originalfile.jpg) and moved into a single directory. It was necessary to maintain source provenance in the filename so that a subsequent deduplication step would be possible.

Stage 3 — Three-tier duplicate removal. Tier 1 used binary matching using MD5 hash, eliminating 8,226 exact matches and showing that source v7 was an exact copy of v6. Tier 2 used perceptual hashing on near-duplicates of JPEG recompression - no more images flagged. Tier 3 compared label files polygon structure of tier 2 to identify augmentation-variant duplicates (the same scene with colour jitter or flipping enabled), eliminating 34 more images. images removed in total: 8,260 images, 14,445 clean annotated images remaining.

Stage 4 — Class ID remapping. A YAML configuration file of each source dataset was read to create a lookup table between the local IDs and the master IDs. All label files were re-written to replace the first token of each annotation line; polygon coordinates were left unchanged. There were 14,435 label files remapped; 10 blank background-only files were deliberately ignored.

Stage 5 — Stratified split rebalancing. The partitioning that was inherited was highly skewed: 96.9% training, 2.8% validation, 0.2% test. A 70/20/10 reshuffle was done on each of the six remaining source datasets independently and then recombined to maintain proportional representation of each of the six source domain sets in all three partitions. Fixed random seed 42 throughout [21].

Stage 6 — Visual quality check. Thirty random images (10 each partition) were programmatically annotated - polygons of segmentation manually drawn on images with labels indicating a class in the center of the polygons - and then manually checked to verify that physical waste objects were correctly delineated by a polygon boundary.

Table 1 is a summary of the entire pipeline outcome of the six stages.

Pipeline Metric	Value
Source datasets merged	7 (v1, v5, v6, v7, nv2, nv3, nv4)
Images before deduplication	22,999
Exact duplicates removed (MD5 hash)	8,226
Dataset v7 — complete duplicate of v6, discarded	Yes
Near-duplicates removed (perceptual hash)	0
Augmentation-variant duplicates removed	34
Total images removed	8,260
Final clean images	14,445
Training images (70%)	10,111
Validation images (20%)	2,889
Test images (10%)	1,445
Number of classes	38
Annotation format	YOLO segmentation polygon
Hazard tiers	LOW / MEDIUM / HIGH

Table 1: Statistics of the data set after the six-stage engineering pipeline.

Assign the structure of classes and level of hazards. The 38 classes are of four groups. High hazard materials include sharps and cutting tools (needle, scalpel, blade, tweezer_metal, tweezer_plastic) and infectious (bloody objects, covid buffer, covid buffer box, covid test case, iodine swab, gauze, cotton swab, bandage). Pharmaceutical packaging and laboratory devices (capsule, pill, drug_packaging, syringe, reagent_tube, reagent_tube_cap, plastic_medical_bottle, medical_infusion_bag, single_channel_pipette, transferpettor_glass, transferpettor_plastic, harris_uni_core, harris_uni_core_cap) are MEDIUM. Protective equipment, and non-hazardous containers (mask, n95, medical_gloves, plastic_medical_bag, paperbox, cap_plastic, glass_bottle, oxygen_cylinder, mercury_thermometer, electronic_thermometer, unguent, radioactive_objects) are LOW or MEDIUM, according to WHO hazardous waste classification guidelines.

4. Methodology

4.1 Objectives and Motivation

The aim is to recognize medical waste, compute a clinically based risk hazard tier of each item, and generate a scene-level risk score of contamination, all in real time, explicit estimates of uncertainty to avoid overconfident choices. The reason is that there is an apparent gap in the current literature: traditional detectors assume all classes are equally valuable and generate point estimates and do not give error bars, which is inadequate when a false classification may lead to a sharp instrument being discarded. This particular problem forms the basis of MMDBHDSNet, whose architecture and a training mission are directly encoded to encode clinical risk priorities.

System pipeline overview. The full process of the end-to-end pipeline, which includes the ingestion of raw frames and two-branch processing, risk scoring, uncertainty estimation, and real-time deployment output, is depicted in Figure 1.

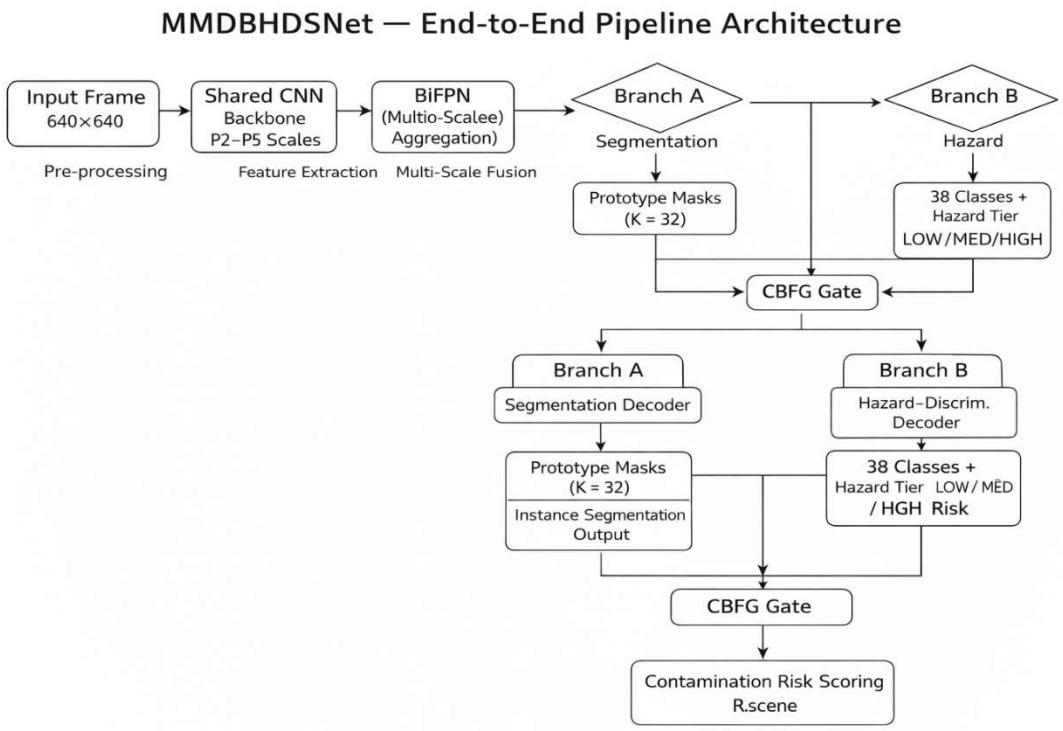

Figure 2: MMDBHDSNet end-to-end pipeline. The input frames are processed with a common BiFPN backbone into parallel segmentation and hazard-discriminative decoders via the Cross-Branch Fusion Gate (CBFG) and

downstream contamination risk scoring and MC Dropout uncertainty estimation before reaching the deployment interface.

4.2 Existing Approaches and Their Limitations

Single-stage detectors like YOLOv8 that are of standard type do not distinguish between classes and have a single classification head. They are good overall detectors but do not have hazard supervision and do not give uncertainty estimates. Mask R-CNN and its successors are accurate but need to compute region proposals, introduce inference overhead. Both architecture do not contain a contamination risk model or direct uncertain predictions to human review. There exist multi-task models, which jointly optimize both segmentation and detection, but which normally share all the parameters of the decoders, which leads to conflicting gradient updates that negatively affect both tasks at the same time [22]. MMDBHDSNet addresses this by isolating branches as well as having a controlled gate of information exchange.

4.3 Proposed Architecture: MMDBHDSNet

Cooperation with BiFPN. The encoder uses a four-resolution scale convolutional feature pyramid on input images with a stride of P2 (stride 4) to P5 (stride 32). The bidirectional feature pyramid network (BiFPN) is a type of cross-scale aggregation using learnable fusion weights to generate enriched multi-scale feature maps, which are shared input to the decoder branches [13]. This provides a combination of fine-grained texture at high resolutions and semantic context at low resolutions in both branches - useful when the detector has to deal with a small pill and a large oxygen cylinder in the same frame.

Branch A — Segmentation decoder. Branch A restores spatial resolution to the original input scale, using transposed convolutional upsampling, and skip connections of the encoder maintain fine spatial detail. Instead of directly predicting masks, it learns K = 32 prototype masks, and a per-instance coefficient vector. Each detection generates a final mask as the weighted sum of these prototypes, which makes it possible to generate masks efficiently during inference time.

Branch B — Hazard-discriminative decoder. Branch B is parallel and shares the same BiFPN feature maps but decodes with different parameters. It places every candidate region in one of 38 waste categories and, at the same time, in one of 3 hazard tiers: LOW (weight 1.0), MEDIUM (2.5), or HIGH (5.0), based on the clinical waste management guidelines and the WHO biosafety categories. The multi-task training loss is $L_total = 1.5 \cdot L_seg + 0.5 \cdot L_cls + 0.3 \cdot L_hazard + 7.5 \cdot L_box$. The loss of each instance is further weighted based on the hazard weight of its ground-truth class, forming a direct gradient incentive to make accurate discrimination of high-hazard items. Focal loss ($\gamma = 2.0$, $\alpha = 0.25$) addresses class imbalance across hazard tiers [23].

Cross-Branch Fusion Gate (CBFG). A Cross-Branch Fusion Gate is added to the P3 and P4 feature scales to allow controlled exchange of information in both directions. With Branch A having h A and Branch B having h B on the same scale, the gate yields $h_A' = h_A \cdot \sigma(MLP([h_A, h_B]))$ and $h_B' = h_B \cdot \sigma(MLP([h_A, h_B]))$, where σ is a sigmoidal and MLP a two layer bottleneck. This enables the segmentation accuracy in Branch A to enjoy hazard context in Branch B and vice versa without sharing gradient paths. This gate introduces about 1.2 M parameters.

Contamination risk scoring. A post-processing module aggregates instance-level detections into a single scene-level risk score: $R_scene = [\Sigma_i\, conf_i \cdot w_h(c_i) \cdot f_spatial(x_i, S)] / N_norm$, where $conf_i$ is detection confidence, $w_h(c_i)$ is the hazard weight of class c_i, and f_spatial amplifies co-located high-hazard items as $f_spatial(x_i, S) = 1 + 0.3 \cdot |\{j \neq i : dist(x_i, x_j) < 200\text{ px}, w_h(c_j) = HIGH\}|$. The score is smoothed using exponential moving average ($\beta = 0.85$). Thresholds: LOW ($R < 0.35$), MEDIUM (0.35–0.65), HIGH (≥ 0.65).

Uncertainty estimation via Monte Carlo Dropout. Dropout layers in the Branch B decoder (retention probability 0.8) remain active at inference time. Ten stochastic forward passes are run per frame; the predictive mean gives the class probability estimate and the predictive variance gives the epistemic uncertainty proxy. Detections with variance above $\tau = 0.05$ are flagged in the deployment interface with a dashed boundary, indicating they require human confirmation before a disposal decision is made [15, 24].

5. Experimental Setup

Hardware. All experiments ran on an NVIDIA RTX 4500 Ada Generation GPU (24 GB GDDR6 ECC VRAM), an Intel Core Ultra 9 CPU (24 cores, 5.1 GHz boost), and 32 GB DDR5 RAM. Mixed-precision (FP16/FP32) training with automatic loss scaling was used throughout.

Training configuration. MMDBHDSNet was trained end-to-end with SGD (momentum 0.937, weight decay 5×10^{-4}) for 300 epochs. The learning rate was initialised at 1×10^{-2} and annealed via a cosine schedule, preceded by a three-epoch linear warm-up. Batch size was 16 with multi-scale training across seven input resolutions from 448 to 832 pixels. Augmentation comprised random horizontal and vertical flips ($p = 0.5$), rotation $\pm10°$ ($p = 0.3$), mosaic composition ($p = 1.0$, disabled in the final 10 epochs), colour jitter, random perspective distortion, and copy-paste augmentation for rare class oversampling [18, 20]. The checkpoint with highest mAP@0.5:0.95 on the validation set was retained.

Evaluation protocol. All metrics are reported on the held-out test partition of 1,445 images using mAP@0.5, mAP@0.5:0.95, overall Precision and Recall at IoU = 0.5, and per-class AP@0.5 for all 38 classes. Detection confidence threshold was 0.25 and NMS IoU threshold was 0.6.

6. Results and Evaluation

6.1 Overall Performance

Table 2 reports aggregate performance on validation and test partitions. On the test set, MMDBHDSNet achieves mAP@0.5 of 0.9474, mAP@0.5:0.95 of 0.7613, precision 0.951, and recall 0.9168. The overfitting gap between validation mAP@0.5 (0.9151) and test mAP@0.5 (0.9474) is 3.23 percentage points — narrow by the standards of multi-class segmentation systems — indicating robust generalisation across the multi-source test distribution. This is attributed to the diverse multi-source training set, the aggressive augmentation pipeline, and multi-scale training, which collectively expose the model to the full range of visual conditions present in the merged dataset [25].

Metric	Validation	Test
mAP@0.5	0.9151	0.9474
mAP@0.5:0.95	0.7282	0.7613
Precision	0.9225	0.9510
Recall	0.8897	0.9168
Overfitting Gap (mAP@0.5)	—	0.0323

Table 2: MMDBHDSNet aggregate performance on validation and test partitions.

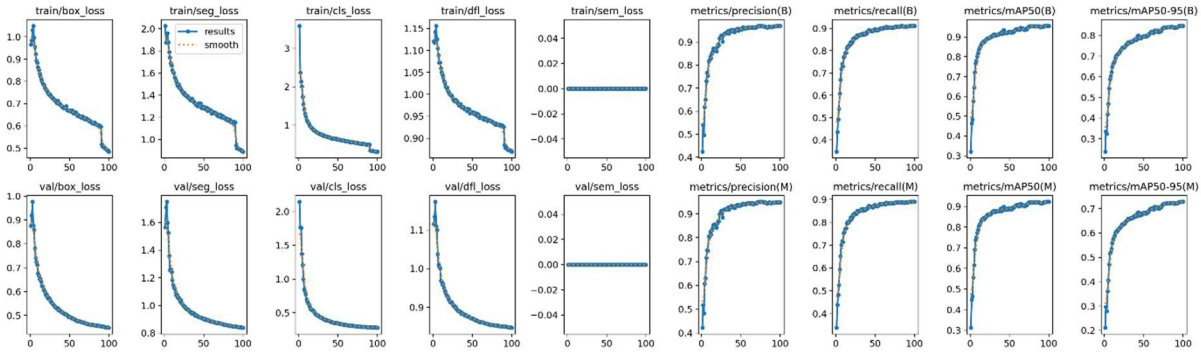

Figure 3: Performance curves of training and validation of MMDBHDSNet versus 300 epochs, with both curves showing consistent convergence and adhering closely to each other during all epochs.

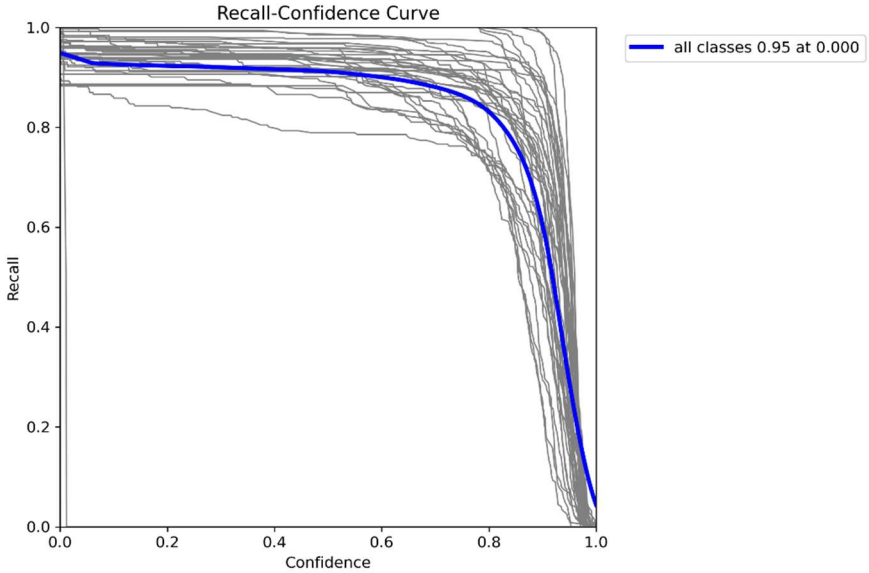

Figure 4: Recall-confidence curve with high recall (approximately 0.95) at lower thresholds with a steep decline at higher levels of confidence, indicating that detection at high levels of confidence is well-calibrated and highly reliable across all classes.

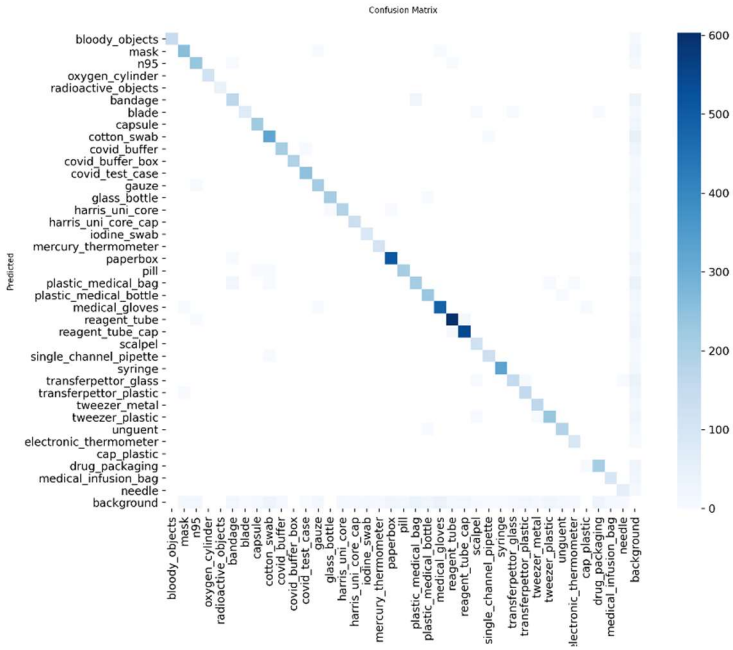

Figure 5: Confusion matrix with an apparent dominance of the diagonal and low off-diagonal errors indicating good classification accuracy among all classes (38) with only some confusion between visually similar or small-object classes.

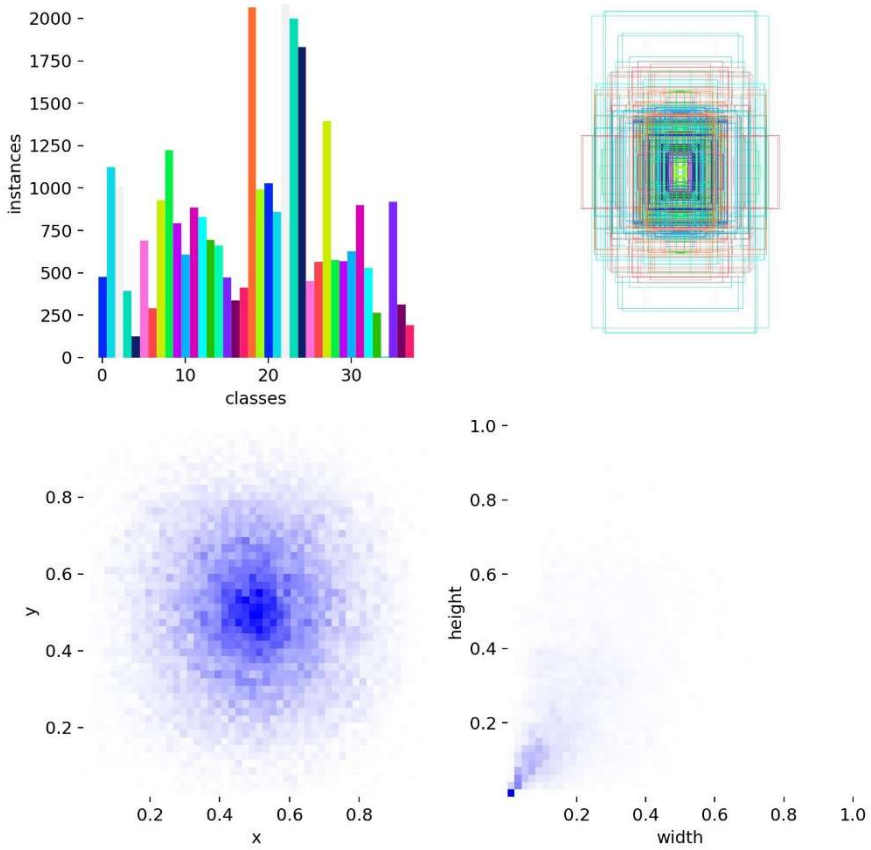

Figure 6: Dataset distribution and spatial analysis showing class imbalance, bounding box overlays, object center density, and width–height distribution, highlighting scale variation and spatial concentration of medical waste instances.

6.2 Per-Class Performance

The AP@0.5 and hazard tier of all 38 classes on the test partition were reported in Table 3. There are three performance bands that are apparent. Classes with distinctive morphology or strong colour cues dominate the top band (AP > 0.97): oxygen_cylinder (0.995), radioactive_objects (0.995), glass_bottle (0.995), covid_buffer_box (0.9948), covid_test_case (0.9943), syringe (0.9944), mercury_thermometer (0.9768), and paperbox (0.9795). The mid-band (0.90–0.97) covers most classes including mask (0.9661), capsule (0.9202), scalpel (0.9521), and medical_gloves (0.9592). The lower band includes needle (0.7269), iodine_swab (0.7826), cotton_swab (0.7931), and blade (0.8285). The worst aspect is that needle and blade have the smallest pixel footprint - often less than 0.2 percent of image area - and are often partially blocked. Cotton swab and iodine swab can be confused with each other due to the only advantageous discriminating variable being the iodine discolouration, which does not appear in unused swabs - a problem of data scarcity, rather than a failure of the model, and the main reason why the active learning approach described in Section 8 was chosen.

Class	AP@0.5	Hazard Tier
bloody_objects	0.9710	HIGH
mask	0.9661	LOW
n95	0.9654	LOW
oxygen_cylinder	0.9950	MEDIUM
radioactive_objects	0.9950	HIGH
bandage	0.9011	MEDIUM

blade	0.8285	HIGH
capsule	0.9202	MEDIUM
cotton_swab	0.7931	MEDIUM
covid_buffer	0.9710	HIGH
covid_buffer_box	0.9948	HIGH
covid_test_case	0.9943	HIGH
gauze	0.9726	MEDIUM
glass_bottle	0.9950	LOW
harris_uni_core	0.9630	MEDIUM
harris_uni_core_cap	0.9582	LOW
iodine_swab	0.7826	HIGH
mercury_thermometer	0.9768	HIGH
paperbox	0.9795	LOW
pill	0.9422	MEDIUM
plastic_medical_bag	0.8955	LOW
plastic_medical_bottle	0.9538	MEDIUM
medical_gloves	0.9592	LOW
reagent_tube	0.9886	MEDIUM
reagent_tube_cap	0.9759	LOW
scalpel	0.9521	HIGH
single_channel_pipette	0.9831	MEDIUM
syringe	0.9944	HIGH
transferpettor_glass	0.9082	MEDIUM
transferpettor_plastic	0.9097	MEDIUM
tweezer_metal	0.9476	MEDIUM
tweezer_plastic	0.9199	LOW
unguent	0.9680	LOW
electronic_thermometer	0.9661	LOW
cap_plastic	0.0000	LOW
drug_packaging	0.9296	MEDIUM
medical_infusion_bag	0.9302	MEDIUM
Needle	0.7269	HIGH

Table 3: AP at 0.5 per-class on all 38 classes of the test partition and hazards tier.

6.3 Hazard-Tier Aggregated Performance

Grouping per-class AP scores by hazard tier reveals: HIGH-tier classes achieve a mean AP@0.5 of 0.921 (excluding cap_plastic); MEDIUM-tier 0.937; LOW-tier 0.951. The small-object categories in that group are hard to classify; this is represented in the lower HIGH-tier mean. Critically, among HIGH-tier classes with adequate training data — syringe (0.9944), covid_test_case (0.9943), covid_buffer_box (0.9948), mercury_thermometer (0.9768), bloody_objects (0.9710) — AP

consistently exceeds 0.97. This establishes that the hazard-weighted loss formulation fulfills its goal: of focusing the capacity of the model on the sets in which being wrong is actually expensive.

6.4 Validation Batch Predictions

Figures 2, 3 and 4 depict sample validation batch prediction grids. A test image is displayed in each grid cell with bounding boxes and class labels model-predicted. The confidence scores of 1.0 or close to it can be seen in most of the correctly identified items which include syringes, medical gloves, n95 masks, paper boxes, mercury thermometers, reagent tubes and transferpettors. The multi-object scenes with the overlapping instances are also correctly handled in the model. Micro errors that were observed in the lower performance band full mix up between cotton swab variants, and some instances of a double-detection on the reagent tube caps are also consistent with the quantitative analysis of Table 3.

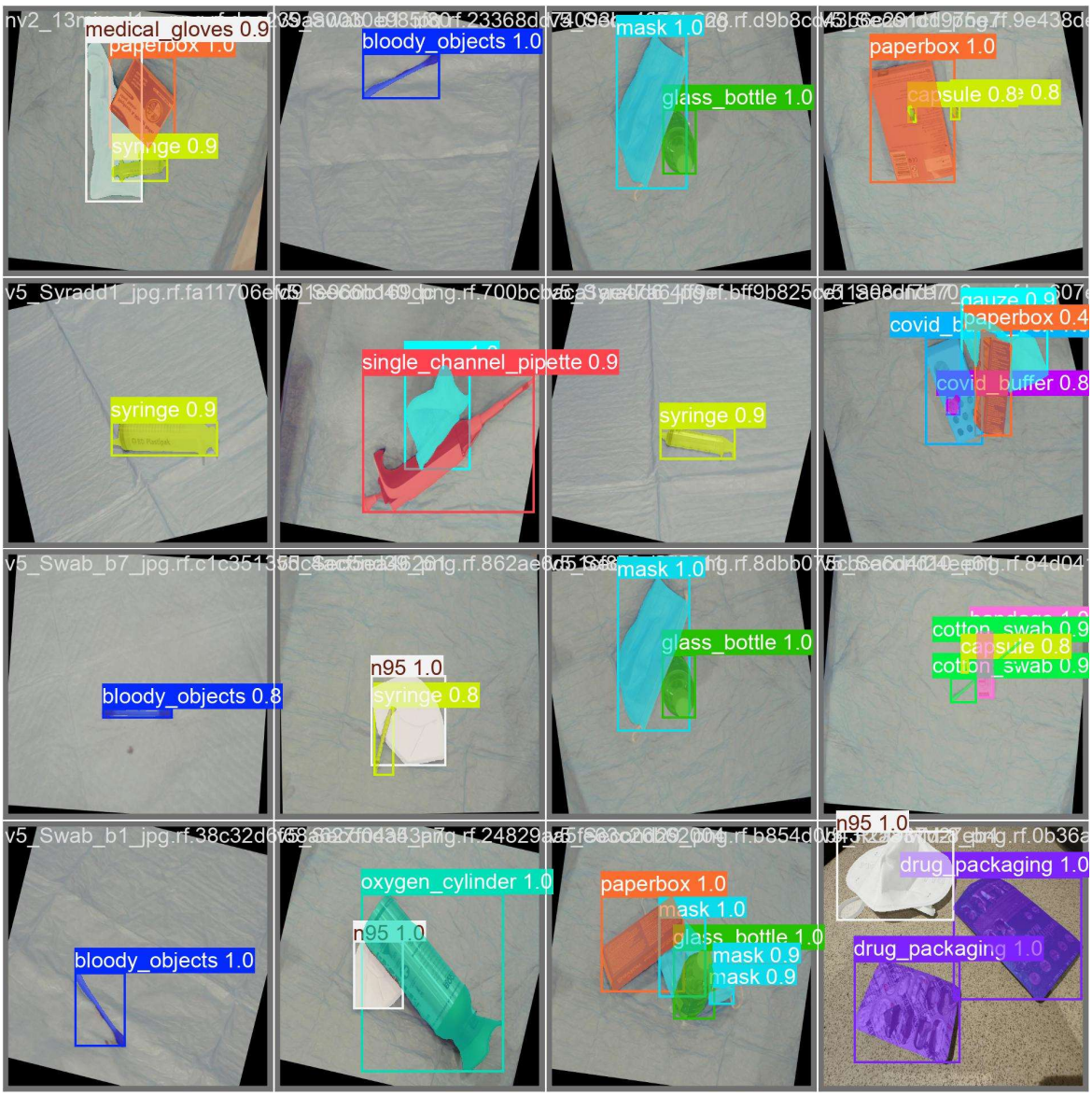

Figure 7: Validation batch 0 - Model predictions on held-out images. The confidence scores and labels are superimposed. Indicate high-confidence findings of syringes, gloves, masks, paperboxes, and bloody objects.

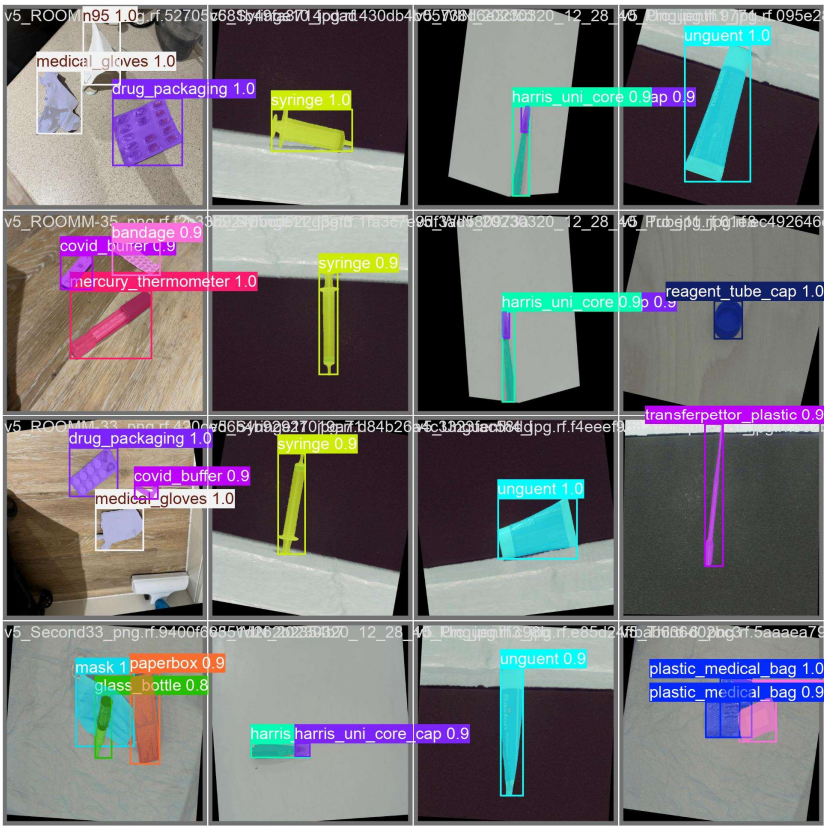

Figure 8: batch 1 validation - Predictions on images of lab scenes. The model accurately recognizes n95 masks, drug packaging, medical gloves, harris core, instruments, reagent tubes, transferpettors, and unguent tubes with confidence rate ≥ 0.9.

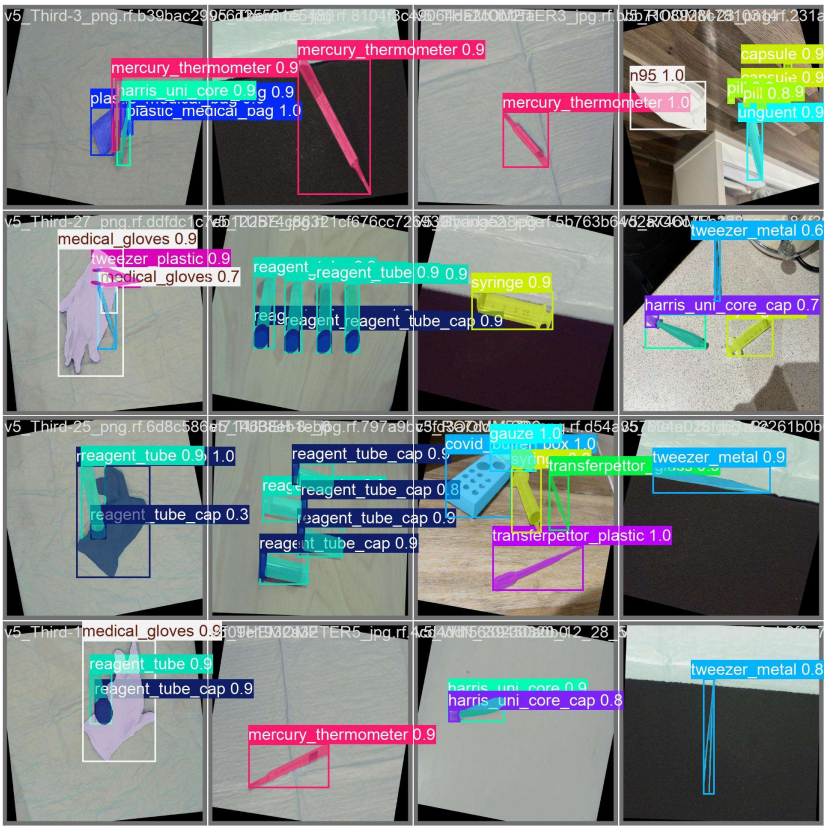

Figure 9: Prediction batch 2 - Predictions on mixed instrument scenes. All mentioned (mercury thermometers, reagent tube rack, tweezer metal, transferpettor plastic, single channel pipettes, harris uni core caps, and plastic medical bags) are accurately identified under different perspectives and lighting.

6.5 Deployment Session Summaries

Figure 10 and Figure 11 display the session summary dashboard of two sample live deployment test sequences with the frequency of class detection, EMA-smoothed risk score over time, and the distribution of hazard level.

Figure 10: Session overview - 2,003 frames. glass_bottle has the most detections (784) then medical_gloves (109) and bandage (100). Hazard distribution: 68.5% LOW, 24.5% MEDIUM, 7.0% HIGH. There are sharp transient spikes (maximum of around 37) at frame 40 and 60 when bloody objects and syringes cooccurred, and EMA decays quickly between events - indicating the correct sensitivity without saturation of false alarms persistently.

Figure 11: Session summary 1,324 frames. plastic_medical_bag (1,091) and plastic medical-bottle (640) are predominant, which is expected in a waste stream of mostly a packaging material. Hazard distribution: 84.7% LOW, 14.7% MEDIUM, 0.6% HIGH. Risk trajectory rapidly decays with a HIGH spike, followed by a stable LOW band throughout the rest of the session.

7. Real-Time Deployment Interface

The inference pipeline executes three parallel threads: a capture thread that obtains frames via USB or IP cameras; an inference thread that implements the entire MMDBHDSNet pipeline (segmentation, classification, uncertainty estimation, risk scoring); and a visualisation thread that produces results with the help of a producer-consumer queue. The system can support about 55 frames per second at 640x640 input resolution (18 ms per frame) on the target hardware. By enabling MC Dropout uncertainty estimation (T = 10 passes), throughput reduces to around 7.7 fps - which is suitable in limiting applications where flagged detections are bypassed to an asynchronous human inspection as opposed to an actual real-time decision.

Identified cases are depicted with semi-transparent filled polygon overlays with hues based on hazard risk: green (LOW), amber (MEDIUM), red (HIGH). Cases where the predictive variance is high have a dashed outline. Spatial density heatmap sums hazard-weighted detections across a temporal rolling 30-second window, to locate areas of sustained high-hazard activity. The risk trajectory graph is an EMA-smoothed view of the last 120 seconds of risk history with the horizontal threshold lines and a

background shading that is updated in real-time to indicate the current tier. The alert escalation system alerts amber banner at MEDIUM risk crossing, auditory sound with red banner at HIGH crossing and automated notification to the facility safety officer when the HIGH risk takes longer than 30 seconds without attention. All events are logged at time to a session audit log to comply with regulations reporting.

8. Future Directions and Conclusion

Future work. The zero AP of cap plastic and below-threshold performance of needle and iodine swab point relate directly to the lack of annotated instances. An active learning loop where the uncertainty estimator flags candidate frames from live clinical feeds for priority annotation will address this iteratively. A large-scale in-situ collection campaign from real hospital waste disposal points will provide distribution-matched data for underperforming categories. Temporal modelling — replacing EMA smoothing with a recurrent or attention-based risk trajectory model — would enable predictive alerts that anticipate HIGH-risk events before they materialise. Architecturally, Vision Transformer backbones offer improved long-range contextual reasoning likely to benefit visually similar class pairs [24, 26]. Knowledge distillation from MMDBHDSNet to a lightweight student network would enable edge deployment on ARM-based smart bin controllers within an integrated facility management system [27].

Conclusion. This paper presented MMDBHDSNet, a dual-branch hazard-discriminative segmentation network that addresses the three structural limitations of prior medical waste recognition systems: insufficient class coverage, absence of hazard semantics, and lack of uncertainty awareness. The Cross-Branch Fusion Gate enables hazard context to sharpen segmentation and vice versa without gradient interference. The hazard-weighted loss ensures that getting a needle wrong, costs the model more than getting a paperbox wrong. Monte Carlo Dropout produces explicit uncertainty estimates that route ambiguous detections to human review. This 38 class benchmark -14,445 highly deduplicated, relabelled and collected images of seven independent sources using a rigorous six stage pipeline- is the most well-documented medical waste segmentation dataset described in the literature. On the held-out test partition, MMDBHDSNet achieves mAP@0.5 of 0.9474, precision 0.951, and recall 0.9168, with a 3.23-point overfitting gap that demonstrates genuine generalisation beyond the training distribution. The system, which is deployed to as much as 55 fps with real time risk visualisation and tiered notifications, is a significant progression towards full automation of AI-assisted clinical waste management to meet the increasing demands of the current healthcare infrastructure.

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